

SparkVault

SparkVault is estate-wide Sparkplug history for Ignition: columnar storage, trend readback through Ignition's own tag-history provider, and plant replay on the equipment's true timestamp.

Perpetual license Ignition 8.1.19+ and 8.3 Runs on: Host

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1. Overview

SparkVault is estate-wide Sparkplug history for Ignition: columnar storage, trend readback through Ignition's own tag-history provider interface, and plant replay. It subscribes to the Sparkplug backbone like any other consumer, so adding history changes nothing at the sites. Values are filed under the equipment's own source timestamp, not the time they arrived, so a site that loses its link and replays its buffer hours later backfills the trend correctly.

Three lanes make up the product: Sparkplug ingest, trend readback out of the database through Ignition's tag-history provider (so existing trend bindings work unchanged), and replay of a stored window back onto the namespace as a synthetic edge.

2. What You Need

- A SparkBridge Host gateway.
- A Postgres/Timescale database reachable from the gateway.
- Ignition's Tag Historian module active and licensed on the gateway (trend readback runs through it).

3. Install

1. SparkVault carries its own license. Install the signed module through Config > Modules > Install or Upgrade a Module; no build step is required.
2. Set `sparkvault.ingest.enabled=true` to activate the ingest lane; it stays inert otherwise.

Schema migrations apply automatically at startup under a Postgres advisory lock.

4. License

See the README in the bundle for licensing detail. Standard suite behavior applies: the key file lives at `<data>/sparkbridge-license.key` and is re-checked every 30 seconds.

5. Configure

<code>sparkvault.ingest.enabled</code>	false by default; master switch for the Sparkplug MQTT ingest lane.
<code>sparkvault.broker.uri</code>	default <code>tcp://127.0.0.1:1883</code> .
<code>sparkvault.db.url</code>	default <code>jdbc:postgresql://localhost:5432/sparkvault</code> .
<code>sparkvault.db.user</code>	default <code>sparkvault</code> .
<code>sparkvault.db.schema</code>	default <code>sparkvault</code> .
<code>sparkvault.jetstream.enabled</code>	false by default; independent JetStream ingest source, stream must be bound to <code>uns.></code> .
<code>sparkvault.write.batch</code>	default 500 rows per insert batch.
<code>sparkvault.buffer.path</code>	default <code>data/sparkvault-history-buffer.dat</code> , the store-and-forward spill file.

Full config key list is in the module README.

6. Operate

SparkVault stores every historized point on the equipment's own source timestamp, not receipt time. On a link outage, the ingest lane spills to `sparkvault.buffer.path` and replays on reconnect, so the trend backfills correctly rather than showing a gap. Any past window can be replayed on demand back onto the namespace as a synthetic edge for analysis tooling that expects live-looking data.

7. Verify

Check `GET /data/io.sparkvault.historian/status.json` for lane state and heartbeat age. Confirm a trend binding against the registered tag-history provider returns data matching the live Sparkplug stream.

8. Troubleshoot

Symptom	Cause	Fix
No data ingesting	<code>sparkvault.ingest.enabled</code> is false (default)	Set it to true and restart or wait for the property re-read
Trend binding returns no data	Tag Historian module trial expired, or the provider is not registered	Reset the Tag Historian trial on the gateway; confirm SparkVault's provider is registered
<code>status.json</code> shows lane not healthy / heartbeat stale	Broker or database connectivity lost	Check <code>sparkvault.broker.uri</code> and <code>sparkvault.db.url</code> reachability from the gateway
Gap in history after an outage that should have replayed	Buffer file missing or not writable	Confirm <code>sparkvault.buffer.path</code> is writable and has not been cleared

9. Reference

- Status route: `GET /data/io.sparkvault.historian/status.json`.

- Default database: `jdbc:postgresql://localhost:5432/sparkvault`, schema `sparkvault`.
- Buffer file default: `data/sparkvault-history-buffer.dat`.
- See also [SparkRecord](#) for structured records and events rather than continuous telemetry.

10. Support

[SparkVault in the docs](#) · [SparkVault on the Host product page](#) · [Contact](#) · [GitHub releases](#)